



**SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL
SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**



COURSE NAME: Theory of Computation

COURSE CODE: CSA1310

COURSE OUTCOME COVERED

CO1: Apply mathematical foundations such as formal proofs, induction, and basic concepts of automata theory to solve computational problems (BL:3)

Assessment Tool Weightage: 10%

QUESTIONS: 5

Total Marks:50

Duration : 1 Hour

Technical Presentation

Code of Conduct:

I Gunapu Sravanthi Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

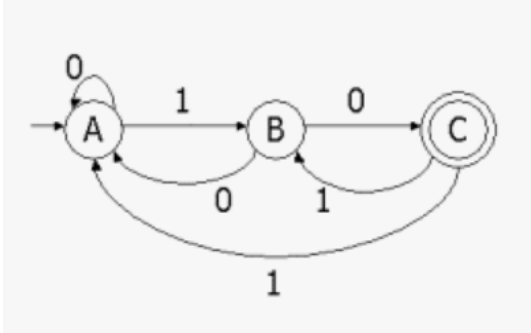
Signature of the student: Sravanthi



SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL
SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



1. How can every Non-Deterministic Finite Automaton (NFA) be converted into an equivalent Deterministic Finite Automaton (DFA), and what are the challenges involved? Consider the NFA given below which accepts some language L over the alphabet $\Sigma = \{0, 1\}$. Construct a DFA that represents the same language L



Simulation:

The screenshot shows the Automaton Simulator web application. The interface includes a top navigation bar with tabs for DFA, NFA, and PDA. The NFA tab is selected. The main area displays the NFA diagram with states A, B, and C. State A is the start state, and state C is the final state. The transitions are: A to A on 0, A to B on 1, B to C on 0, C to B on 1, B to A on 0, and C to A on 1. The left sidebar contains a 'Test / Debug' section with a 'Bulk Testing' button and a 'Test Results' section. The 'Test Results' section shows the following results:

- Accept: 0 -- Pass
- Accept: 00 -- Pass
- Accept: 100 -- Pass
- Reject: 1 -- Pass
- Reject: 0 1 -- Pass